

Product Specification

SenseAir[®] *S8-4BP*

Miniature CO₂ sensor safety switch

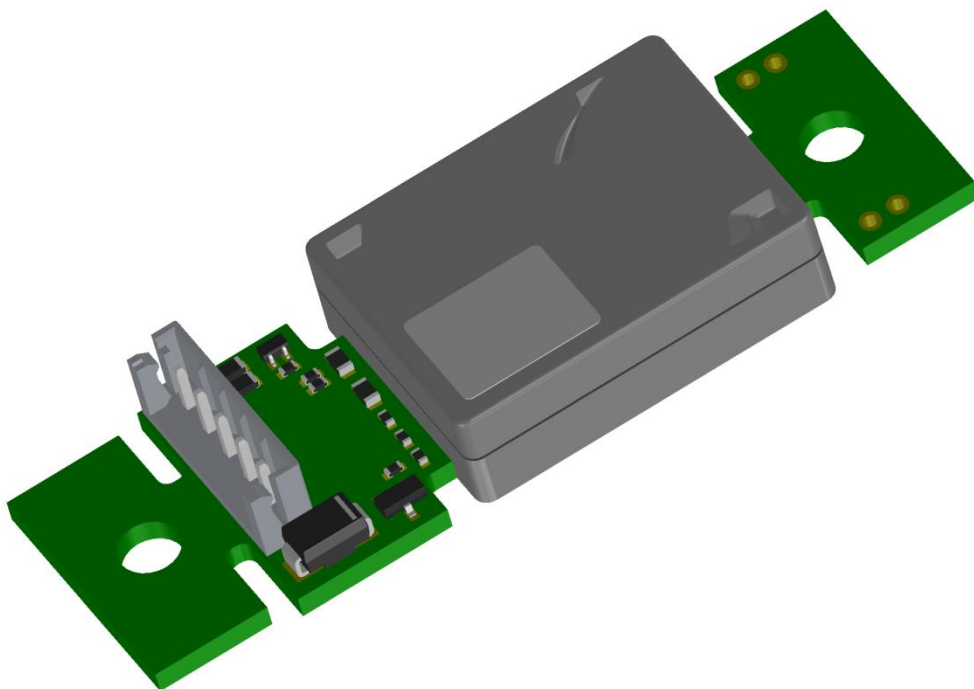


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1. General

The *SenseAir® S8-4BP*, Article number 004-0-0023, CO₂ sensor module is designed to serve as a CO₂ safety switch when built-in into kerosene heaters. The sensor utilizes reliable and highly accurate infrared gas sensing technology. The electronic circuitry is optimized for low power consumption.

The design of *SenseAir® S8-4BP* will comply with or exceed specifications for *CO₂ ENGINE™ model-4BP* edition 6 dated 2005-12-12.

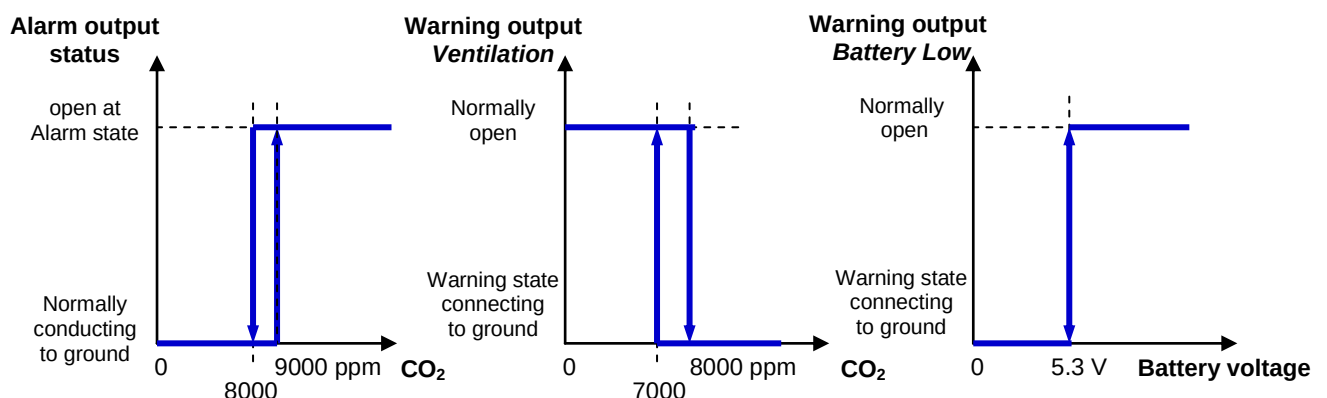
2. S8-4BP functional description

During normal operation, the sensor measures ambient gas CO₂ concentration every 30 seconds and will set alarm output when CO₂ level is higher than 9000 ppm. A diagnostic routine will set *Fault Alarm* if any malfunction is detected. An alarm filter protects the sensor from issuing false alarm caused by intermittent short disturbances.

The alarm output is an open drain transistor switch which sinks the output pin to zero volts in operation mode and alarm is represented by open state (i.e. non-conducting). If any of the three alarm states CO₂ High, Power Low or Fault occur the unit sets the alarm. The output will stay in alarm mode until CO₂ level is below 6500ppm or other reason for alarm has been removed.

The SenseAir® S8 model-4BP features, in addition to the main alarm output, one warning output for Battery Low indication and one warning output for Ventilate indication. Warning outputs are capable to drive an external LED indicator.

The Ventilate indication warns when the CO₂ level exceeds 8000 ppm. The warning outputs are to alert the user that the heater may soon switch off automatically.



3. Key technical specification

Item	<i>SenseAir[®] S8-4BP</i>
Target gas	CO ₂
Operating Principle	Non-dispersive infrared (NDIR)
Measurement range	400 to 32000ppm (represented internally in digital format)
Accuracy	±1000ppm at alarm points between 7000 and 9000 ppm (Notes 1)
Gas diffusion response time	2 minutes by 90%.
Operating temperature	-5° to 60° C
Operating humidity range	0 to 95% RH non condensed
Storage temperature	-40° to + 70° C
Storage Environment	0 to 95% RH non condensed non corrosive gases no contamination to kerosene
Dimensions (mm)	60 x 19.6 x 7.6
Weight	< 10 grams
Power supply	4.5 to 7.0 VDC maximum rating (without reverse polarity protection)
Power consumption	250 mA peak, 2 mA average
Life expectancy	5+ years in normal indoor / office environments
Compliance with	Tested according Emission: EN 61000-6-3:2007, EN 61000-6-4:2007 Immunity: EN 61000-6-1:2007 RoHS directive 2002/95/EG
Output Alarm, Open Drain	9000/8000 Normal state is conducting, max 100 mA. Transistor stops sink output at (CO ₂ >9000ppm) OR (Unloaded Power voltage < 4.5V) OR (Loaded Power voltage < 4.0V) OR (Sensor Failure is detected in self-diagnostics).
Battery Low Warning, Open Collector	Built in series 390R resistor for direct LED driving Normal state is non-conducting Transistor becomes conducting below <i>Battery Low CO₂</i> set point (5.3V).
Ventilate Warning, Open Collector	Built in series 390R resistor for direct LED driving 8000/7000 Normal state is open. Transistor becomes conducting above <i>CO₂ Level Pre-Alarm</i> set point.
Maintenance	Forced calibration (assuming 400 ppm exposure).
Forced alarm test pad	Provides outputs activation for test purposes
Self-diagnostics	Full self-diagnostics at power up and continuously running self-diagnostics at every measurement.

Table I. Key technical specification for the *SenseAir[®] S8-4BP*

Note 1: Accuracy is specified over operating temperature range. Specification is referenced to certified calibration mixtures. Uncertainty of calibration gas mixtures (+/-2% currently) is to be added to the specified accuracy for absolute measurements.

4. Pin assignment

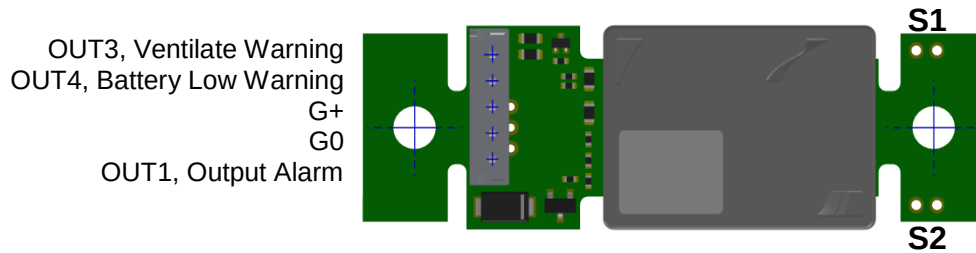


Figure 1. Pin assignment *SenseAir® S8-4BP*.

5. Terminals description

The table below specifies terminals and I/O options dedicated in *SenseAir® S8-4BP* model.

Pin Function	Pin description / Parameter description	Electrical specification
Power supply		
G0	Power supply negative terminal. Sensor's reference (ground) terminal.	Unprotected against reverse connection!
G+	Power supply positive terminal.	Unprotected against reverse connection!
Outputs		
Output Alarm	Open Drain output for alarm indication. Internal protection.	max 100 mA, pull up by 100k to G+
Warning output Ventilation	No internal protection, Pulled up to G+ at processor reset (power up and power down)	Built in series 390R resistor for direct LED driving
Warning output Battery Low	No internal protection, Pulled up to G+ at processor reset (power up and power down)	Built in series 390R resistor for direct LED driving

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Jumpers		
S1 Calibration restore switch	Digital input forcing background calibration. Background calibration is activated when closed for minimum 60 sec assuming 400 ppm CO2 sensor exposure. Calibration occurs every 60 sec during switch grounding (Note 2) Absolute max voltage range(Note 1) Internal pull up resistor Input low level (Note 1) Input high level (Note 1)	No internal protection, Internal pull-up to 3.3V at processor reset (power up and power down) - 0.3V to 3.8V 120K - 0.3V to 0.75V 2.3V to 3.6V
S2 Forced Output test	Digital input enabling outputs test mode (when closed for minimum 30 sec). Absolute max voltage range(Note 1) Internal pull up resistor Input low level (Note 1) Input high level (Note 1)	No internal protection, Pulled up to 3.3V at processor reset (power up and power down) - 0.3V to 3.8V 120K - 0.3V to 0.75V 2.3V to 3.6V

Table II. I/O notations, description and electrical specification.

Note 1: Specified parameter relies on specification of subcontractor and is not tested by SenseAir

Note 2: .Do not ground bCal input for long time. FLASH resource will be exhausted in 3.5 months in case of permanent bCal grounding.

6. Gas diffusion area

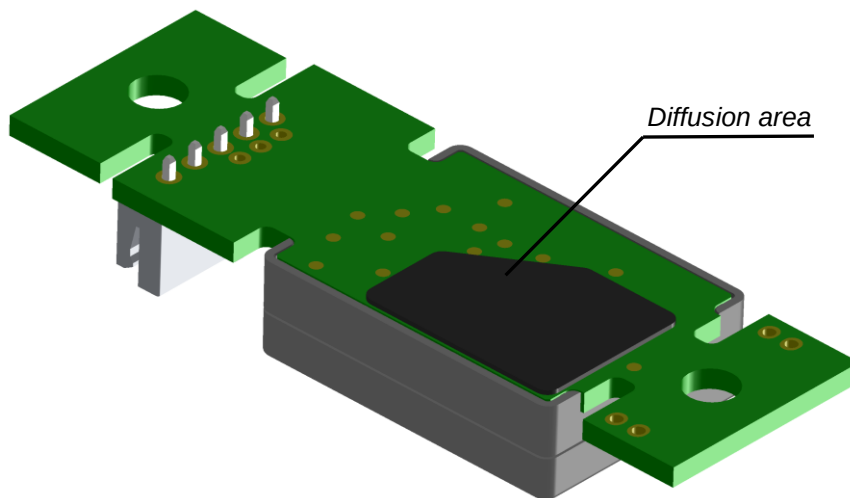


Figure 2

7. Absolute maximum ratings

Stress greater than those listed in Table III may cause permanent damage to the device. These ratings are stress ratings only. Operation of the device at any condition outside those indicated in the operational section of these specifications is not implied. Exposure to absolute maximum rating for extended periods may affect device reliability.

Parameter	Minimum	Maximum	Units	Notes
Ambient temperature under bias	- 40	85	C	
Voltage on G+ pin with respect to G0 pin	- 0.3	12	V	1
Maximum output current from active output pin	- 25	+ 25	mA	1
Maximum current on input	- 5	+ 5	uA	1
Maximum voltage on Calibration restore(S1), Forced output test(S2) inputs	- 0.3	3.8	V	1
Maximum voltage on Alarm Out	- 0.3	G+ + 0.5	V	1,2

Table III. Absolute maximum ratings specification for the *SenseAir® S8-4BP*

Note 1: Specified parameter relies on specification of subcontractor and is not tested by SenseAir

Note 2: OUT1 (Output Alarm) pin is internally pulled up to G+. External pull up to higher voltage will provide resistive divider powering sensor via high resistance.

8. Mechanical properties

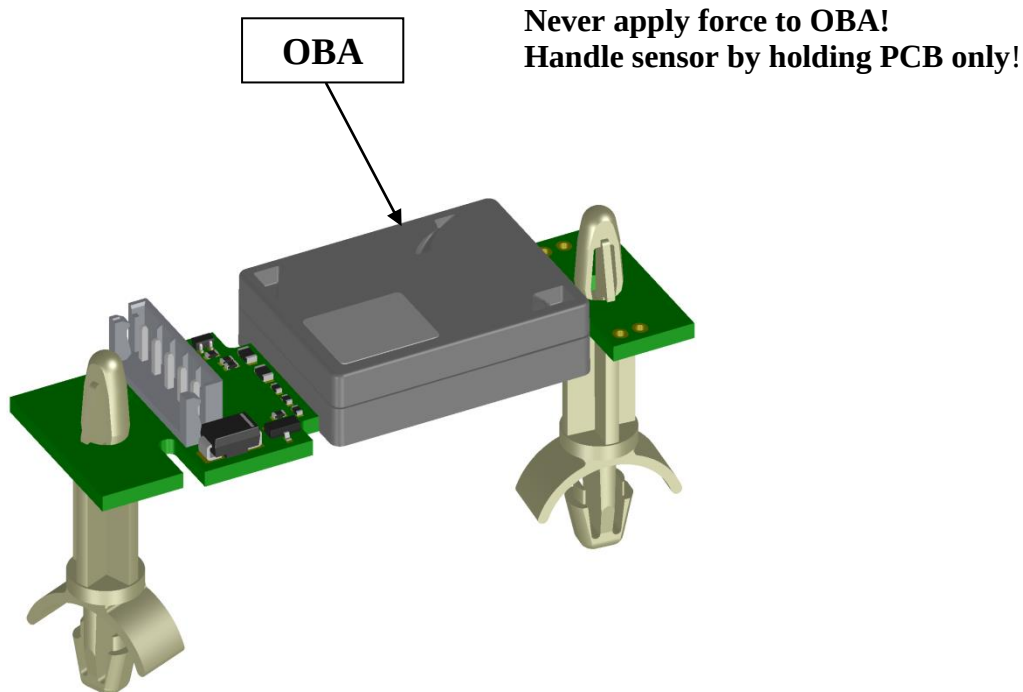
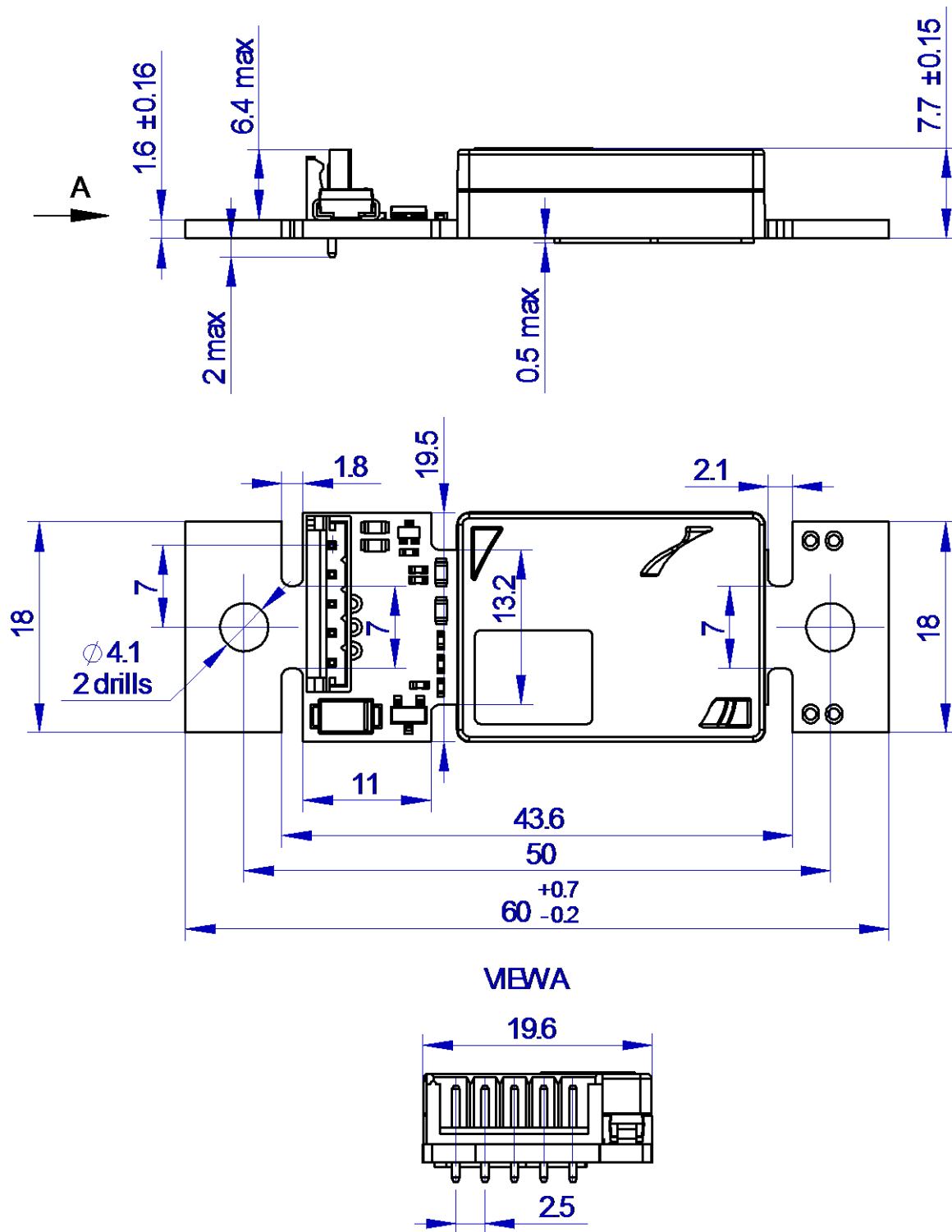


Figure 3. Mechanical drawing *SenseAir® S8-4BP*

Sensor PCB may be colour green or black. Optical bench assembly (OBA) may be colour silver or black. PCB may be fitted with 3, 4 or 5 pin connector. The PCB is fitted with holes for PCB spacers to facilitate attachment to equipment. Please refer to figure 4 for mechanical drawing and detailed specification of dimensions and tolerances.

WARNING!

Under no circumstances apply any force to the OBA, this may permanently harm the sensor and most definitely affect performance. Sensor should be handled holding PCB only

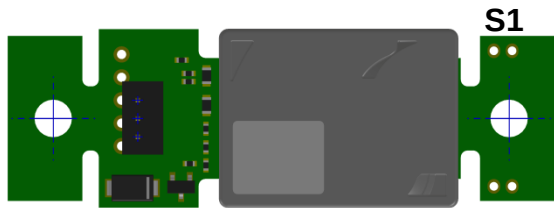


Note: unspecified tolerances are ± 0.1 mm

Figure 4. Mechanical drawing *SenseAir*[®] S8-4BP

9. Maintenance

9.1 Calibration restore switch S1



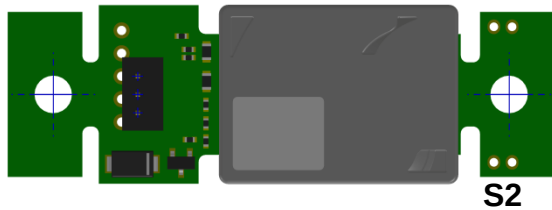
If for some reason the sensor needs to be re-calibrated, this is possible to do by a qualified operator, provided that the sensor is exposed to fresh air during the whole process (~400 ppm CO₂).

The process is actuated by creating an electrical short-cut between the two holes labelled S1. As soon as the micro-controller detects this manually shorted switch terminal S1, calibration is restored to fresh air concentration value.

The delay between the shorting of the switch contact S1 and the actual calibration may be up to 30 seconds

If the operator keeps the sensor with S1 closed for some period of time, the sensor will continue to recalibrate fresh air concentration target value every 30 seconds, until the switch is released.

9.2 Test mode for customer



Forced output test switch S2

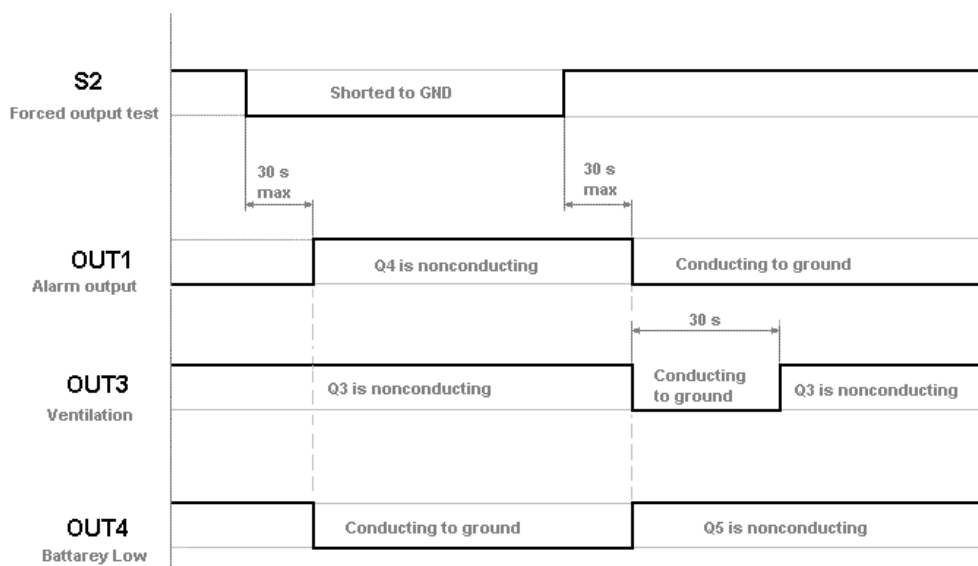
This function is intended for a qualified operator to check the sensor output and the subsequent system response by simulating sensor alarm.

The process is triggered by putting a short across S2. The delay between shorting S2 and setting of alarm active may be up to 30 seconds.

When the micro-controller detects S2 is shorted the Alarm Output and Battery Low Warning is set then after removing the short from S2 the Ventilation Output Warning is set. In an installed sensor this sequence might be hard to detect, since normally the Kerosene heater will power down when S2 is shorted.

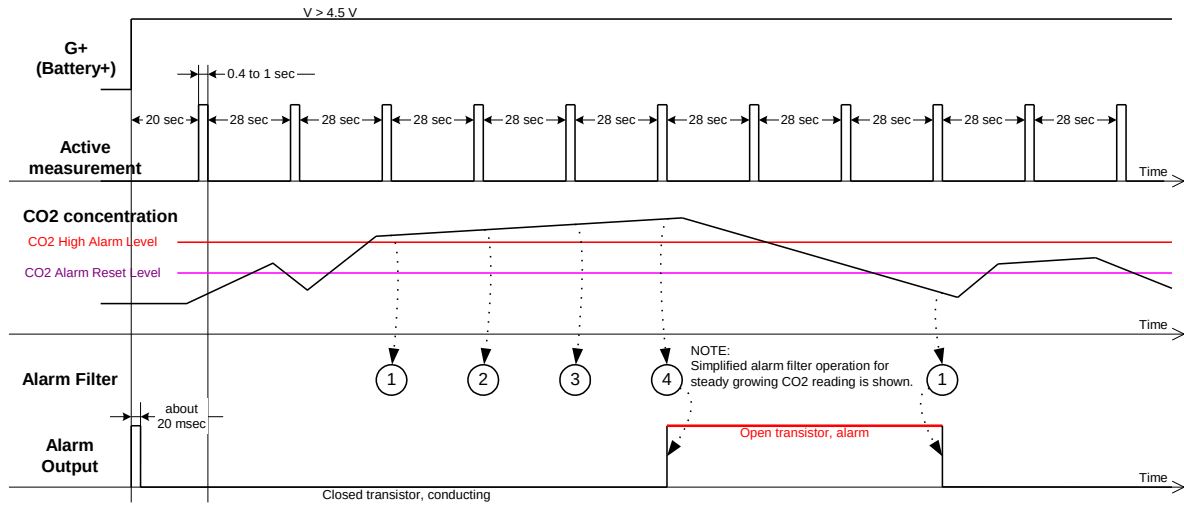
If S2 remains closed for some period of time, and if sensor is powered, the sensor will hold the output active until the switch closure is released. Delay of up to 30 seconds may occur between switch release and alarm output release. Delay of up to 30 seconds may occur between switch release and alarm output release. Ventilation Warning alarm is set until the next measurement cycle after about 30 seconds.

S8-PGA-4BP Outputs test mode

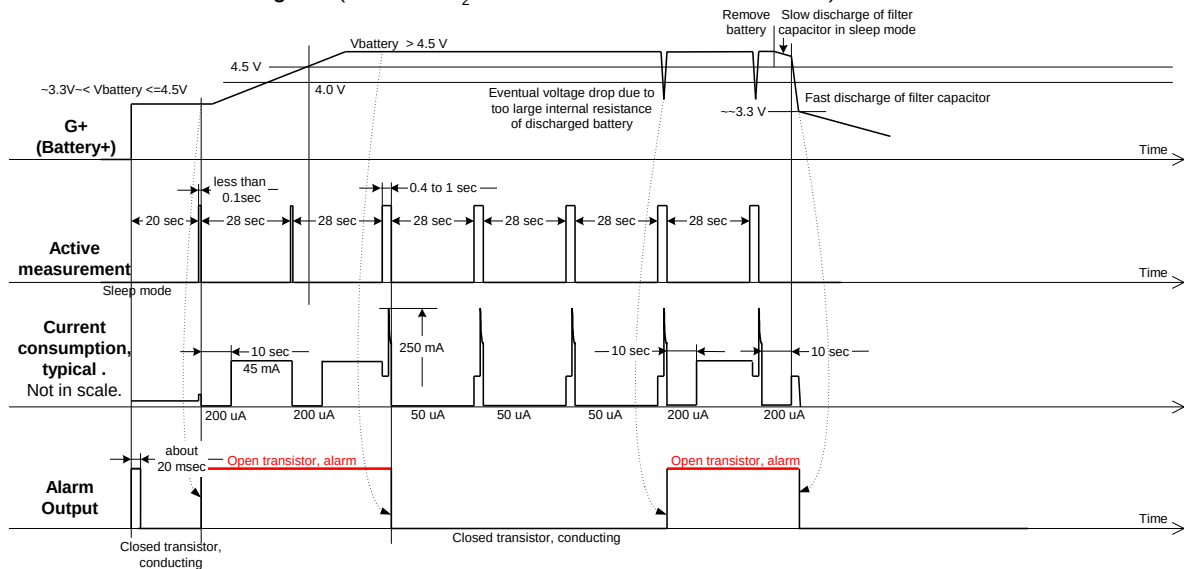


10. Alarm output filter and time diagram

1. High CO₂ alarm time diagram. (Assume good power).



2. Power Low time diagram. (Assume CO₂ concentration to be under alarm level).



Gas and Air Sensors



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